

Team-Aytche BucStop Snapshot

This document will serve as a snapshot in time from the last week of April 2025 of the project **BucStop** and the progress/direction taken by Team Aytche. Upon completion of reading this document, the developer, product owner, or Scrum Master should have a basic understanding of moving parts within the project, and should know how to run the project through Docker.

The main directory of BucStop contains:

- BucStop_APIGateway
- BucStop_GameSubmission
- BucStop_Pong
- BucStop_Snake
- BucStop_Tetris
- BucStop_Webapp

Team Aytche primarily focused on the Webapp directory, as well as the GameSubmission directory. APIGateway, Pong, Snake, and Tetris were not touched, and came from previous teams implementation. Parts of these, especially the APIGateway, are not being used but are being saved for when their implementation may be more viable.

BucStop_Webapp:

Inside of the BucStop_Webapp folder, we have the Documents folder, the BucStop main folder, and scripts. We primarily have not used scripts. Inside of the documents folder, we have documents explaining deployment on Akamai (which is where Team Aytche opted to deploy.) We also have included documentation on Microservices, Metrics that would be useful to report, and how to run locally. The Akamai deployment and local deployment contain the docker information necessary to get this project up and running.

Inside of the BucStop folder, is where the heart of the actual web application users view is. Inside of this folder, we have the Dockerfile to actually run the web application, as well as several important folders. Most notable, inside of MicroServices, we have previous team's implementation of microservices throughout the BucStop web application. Another important folder is the Views folder. Inside of it, the Home views can be seen. We modified the GameCriteria html to allow for GameSubmissions to be done with an API, as well as use an actual web form instead of a form through Google Forms.

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Inside of the BucStop folder, you also have the Controllers Folder. These controllers are vital for existing services. The main challenge of the semester was moving the game submission controller outside of this webapp directory and into its own directory (and subsequently its own docker container.)

BucStop_GameSubmission:

Inside of the GameSubmission Folder, we successfully were able to pull logic from existing microservices to this one. This one has its own Docker file that can be used to launch this microservice inside of its own container. The webapp does not rely on this container being ran, just like how this container can run without the web application. The semester ended as we were wrapping up having the containers communicate with eachother. This container can be launched with the same way as the other commands in the two deployment documents.

The final file I will discuss is the docker-compose.yml file found in the main directory of the GitHub repository. This compose file will allow you to deploy in one command. It will launch both containers, as well as link them to Docker volume (this volume may need to be created manually.)

The main branch structure existed as $\text{prod} \leftarrow \text{test} \leftarrow \text{dev}$

Prod is the production environment, and the code from prod every sprint was ran through Docker to be made public. The test environment is where everyones dev branches were uploaded to, and where any merge conflicts were fixed and tested. The dev branches were per individual, and all uploaded to test prior to prod.

Lastly, after every sprint, a new branch was created called prodXX where XX is the date. This was to ensure a branch could be rolled back to if needed. There is also a branch called local-logging, which is around the same as the 24apr production environment. The difference is this version will not run in a deployment environment, but has logging metrics introduced throughout running inside of Visual Studio.